**National Computing Education Accreditation Council**
NCEAC

NCEAC.FORM.001-D

# COURSE OUTLINE

INSTITUTION **University of Management & Technology, Lahore**

## Course Description:

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|---|---|
| **Course Code** | CS341 |
| **Course Title** | Theory of Automata (TOA) |
| **Credit Hours** | 3 |
| **Prerequisites by Course(s) and Topics** | None |
| **Assessment Instruments with Weights** (homework, quizzes, midterms, final, programming assignments, lab work, etc.) | • Home Assignments + Class Participation **10%**<br>• Quizzes (Approx. 8) **25%**<br>• Mid / Sessional Exam **25 %**<br>• Final Exam **40%** |
| **Course Moderator** | Waleed Riaz |
| **Current Catalog Description** | https://lms.umt.edu.pk/course/ |
| **Textbook** (or **Laboratory Manual** for Laboratory Courses) | Introduction to Languages and the Theory of Computation, by John C. Martin, 4th or later edition. |
| **Reference Material** | • Automata, Computability and Complexity: Theory and Applications, by Elaine Rich, 2011<br>• An Introduction to Formal Languages and Automata, by Peter Linz, 4th edition, Jones & Bartlett Publishers, 2006<br>• Theory of Automata, Formal Languages and Computation, by S. P. Eugene, Kavier, 2005, New Age Publishers<br>• Introduction to Computer Theory, Daniel I. A. Cohen, 2nd Edition |
| **Course Goals** | Theory of automata is a course to explore the theoretical foundations of computer science. In this course, students learn to classify machines by their power to recognize languages from the perspective of formal languages. The aim of this course includes:<br>a) Introduction of the basic theory of Computer Science and formal methods of computation like automata theory, formal languages and grammars<br>b) differentiate between regular, non-regular and context-free languages.<br>c) acquire concepts of computational theory and models such as automata and Turing Machines. |

## Course Learning Outcomes (CLOs):

| Measurable Learning Outcomes | CLOs | Description | Domain & BT Level * |
|---|---|---|---|
| | CLO 1 | **Explain and differentiate** the different concepts in automata theory and formal languages | C1 (Knowledge)<br>C2 (Understand) |
| | CLO 2 | **Prove** properties of languages, grammars and automata with rigorously formal mathematical methods | C5 (Evaluate) |
| | CLO 3 | **Design** of FA, RE, CFG, PDA, and TM | C6 (Create) |
| | CLO 4 | **Transform** between equivalent NFAs, DFAs and REs | C3 (Apply) |
| * BT= Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A= Affective domain | | | |

## Mapping of CLOs to Program Learning Outcomes (PLOs):

| CLOs/PLOs | CLO 1 | CLO 2 | CLO 3 | CLO 4 |
|---|---|---|---|---|
| **PLO 1: Academic Education** | ✔ | | | |
| **PLO 2: Knowledge for Solving Computing Problems** | | ✔ | ✔ | ✔ |
| PLO 3: Problem Analysis | | | | |
| PLO 4: Design and Development of Solutions | | | | |
| PLO 5: Modern Tool Usage | | | | |
| PLO 6: Individual and Teamwork | | | | |
| PLO 7: Communication | | | | |
| PLO 8: Computing Professionalism and Society | | | | |
| PLO 9: Ethics | | | | |
| PLO 10: Life Long Learning | | | | |

| **Tentative Course Plan** | | | | |
|---|---|---|---|---|
| **Week** | **Topics** | **Chapter** | **Assessment** | **CLOs** |
| 1 | **Introduction**<br>Introduction to Theory of Automata, Introduction to Formal Languages, Finite vs. Infinite Languages, Operations performed on strings, Kleene Closure property | Ch 2 and Ch 3 | | **CLO** 1, |
| 2 | **Finite Automata (FA)**<br>Introduction and defining FA, Rules of constructing a FA, Realizing the acceptance and rejection of strings | Ch 5 | Assignment 1 | **CLO** 1, **CLO** 3 |
| 3 | **Finite Automata (FA)**<br>Constructing Transition Table, Relationship of FA with RE, Introduction to NFA, DFA vs. NFA, NFA to DFA conversion, related examples | Ch 5 | Quiz 1 | **CLO** 1, **CLO** 3, |
| 4 | **Regular Expression (RE)**<br>Introductions to RE, strings generation from RE | Ch 4 | Quiz 2 | **CLO** 1, **CLO** 3 |
| 5 | **Transition Graphs (TG)**<br>Introduction to TG and Generalized (TG), Difference between TG and GTG, Examples of TG and GTG, Introduction to Non deterministic Finite Automata (NFA) | Ch 6 | Assignment 2 | **CLO** 1, **CLO** 3, |
| 6 | **Kleen's Theorem**<br>Introduction to Kleene's Theorem, turning TG into RE, Converting RE to FA, Unification | Ch 7 | Quiz 3 | **CLO** 1, **CLO** 2, **CLO** 3, **CLO** 4, |
| 7 | **Finite Automata with Output**<br>Moore Machines, Mealy machines, Moore=Mealy, Revision of topics for Mid Term | Ch 8 | | **CLO** 1, **CLO** 3, **CLO** 4, |
| 8 | **Mid –Term** | | Midterm Exam | |
| 9 | **Regular Languages (RL)**<br>Introduction to Regular Languages, Closure Properties of Regular Languages (R.L), Product, Intersection, Complement of RL | Ch 9 | | **CLO** 1, **CLO** 3, **CLO** 4, |
| 10 | **Non-Regular Languages**<br>Introduction to Non regular Languages, Proving the Non-regularity using Pumping Lemma property | Ch 10 | Quiz 4 | **CLO** 1, **CLO** 2, |
| 11 | **Context Free Grammars (CFG)**<br>Introduction to CFG and CFL, Generative Grammars, Derivations (Left most and right most derivations), Parse Tree/ Syntax Tree, Ambiguity, Total Language Tree | Ch 12 | Assignment 3 | **CLO** 1, **CLO** 3, |
| 12 | **Grammatical Format**<br>Killing $\Lambda$ productions, Killing Unit Productions, Chomsky Normal Form (CNF) | Ch 13 | Quiz 5 | **CLO** 3, **CLO** 1, |
| 13 | **Push Down Automata (PDA)**<br>Introduction to PDA, Pushdown Stack<br>● DPDA vs. NPDA<br>● How the strings are parsed using PDA | Ch 14 | Assignment 4 | **CLO** 1, **CLO** 3, |
| 14 | **CFG =PDA**<br>Constructing a PDA for different CFG, Difference between PDA and FA | Ch 15<br>Ch 11 | | **CLO** 1, **CLO** 4, |

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|---|---|---|---|---|
| | **Decidability**<br>Equivalence and finiteness | | | |
| 15 | **Turing Machines (TM) Single Tape, Mutli Tape**<br>Introduction to TM, Complexity issues and analysis, P and NP problems, Revision of the course and problems discussion | Ch 19 | Quiz 6, | **CLO** 1,<br>**CLO** 3, |
| 16 | Final | | Final Exam | |

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|---|---|---|---|---|
| Laboratory Projects/Experiments Done in the Course | None | | | |
| Programming Assignments Done in the Course | None | | | |
| Class Time Spent on (in credit hours) | 3 hours per week | | | |
| | | | | |
| Oral and Written Communications | | | | |

*-Tentative

## Mapping of CLOs to Direct Assessments

| CLO | Quiz-1 | Quiz-2 | Quiz-3 | Quiz-4 | Quiz-5 | Quiz-6 | A-1 | A-2 | A-3 | A-4 | Midterm | Final |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | ✓ | | | | | | ✓ | | | | ✓ | ✓ |
| 2 | | | | ✓ | | | | | ✓ | | | ✓ |
| 3 | | ✓ | | | ✓ | ✓ | | ✓ | | ✓ | ✓ | ✓ |
| 4 | | | ✓ | | | | | | | | ✓ | ✓ |

Instructor Name:

Instructor Signature ______________________________